

B.Tech. Degree V Semester Special Supplementary Examination September 2022

ME 19-205-0502 MECHANICS OF MACHINERY
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the principles of kinematic pairs, chains and their classification, DOF, inversions, equivalent chains and planar mechanisms.

CO2: Analyze the planar mechanisms for position, velocity, and acceleration.

CO3: Synthesize planar four bar and slider crank mechanisms for specified kinematic conditions.

CO4: Evaluate gear tooth geometry and select appropriate gears for the required applications.

CO5: Design cams and followers for specified motion profiles.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

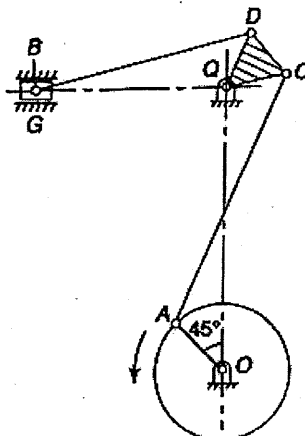
(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	What is a kinematic chain and a kinematic pair?	3	L1	1	1
	(b)	Define mechanical advantage. At what condition mechanical advantage becomes infinite.	3	L1	2	1,2
	(c)	What is meant by backlash in context to gears?	3	L1	4	1,2,3
	(d)	What is the difference between simple gear train and epicyclic gear train?	3	L1	4	1,2,3
	(e)	Define pressure angle in a cam?	3	L1	5	1,2,3
	(f)	What is meant by self locking of screws?	3	L1	3	1,2,3
	(g)	Explain the working of a belt transmission dynamometer.	3	L1	3	1,2,3
	(h)	What do you mean by synthesis of a mechanism?	3	L1	3	1,2,3

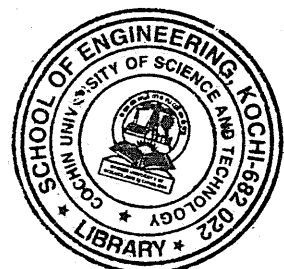
PART B

(4 × 12 = 48)

- II. (a) Explain with examples inversions of a slider crank mechanism. 6 L2 1 1
- (b) The lengths of various links of a mechanism shown in figure are as follows: OA = 150 mm, CD = 125 mm, BD = 500 mm, OQ = 625 mm, AC = 600 mm, CQ = QD = 145 mm. The crank OA rotates at 60 rpm in the counter-clockwise direction. Determine the velocity of the slider B and the angular velocity of the link BD when the crank has turned through an angle of 45° with the vertical.



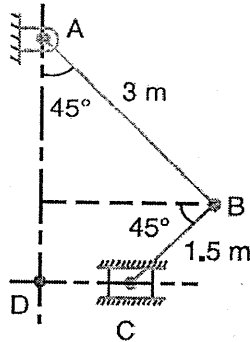
OR



(P.T.O)

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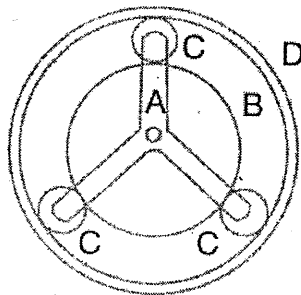
- | | Marks | BL | CO | PO |
|--|-------|----|----|-----|
| III. In the mechanism shown in figure, the slider C is moving to the right with a velocity of 1 m/s and an acceleration of 2.5 m/s^2 . The dimensions of various links are $AB = 3 \text{ m}$ inclined at 45° with the vertical and $BC = 1.5 \text{ m}$ inclined at 45° with the horizontal. Determine: | 12 | L3 | 2 | 1,2 |
| (i) the magnitude of vertical and horizontal components of acceleration of the point B, | | | | |
| (ii) the angular acceleration of the links AB and BC. | | | | |



- | | | | | |
|---|---|----|---|-------|
| IV. (a) State and prove law of gearing. | 4 | L2 | 4 | 1,2,3 |
| (b) Determine the least number of teeth that can be used on each wheel, when Gear ratio is 4:1. Pressure angle is 14.5° . The arc of approach should not be less than circular pitch. The teeth are of involute profile and smaller wheel is the driver. | 8 | L3 | 4 | 1,2,3 |

OR

- | | | | | |
|--|----|----|---|-------|
| V. In an epicyclic gear train of the 'sun and planet' type shown in figure, the pitch circle diameter of the internally toothed ring is to be 224 mm and the module 4 mm. When the ring D is stationary, the spider A, which carries three planet wheels C of equal size, is to make one revolution in the same sense as the sunwheel B for every five revolutions of the driving spindle carrying the sunwheel B. Determine suitable numbers of teeth for all the wheels. | 12 | L4 | 4 | 1,2,3 |
|--|----|----|---|-------|



(Continued)

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		Marks	BL	CO	PO
VI.	The following data relate to a cam profile in which the follower moves with uniform acceleration and deceleration during ascent and descent. Minimum radius of cam = 25 mm Roller diameter = 7.5 mm Lift = 28 mm Offset of follower axis = 12 mm towards right Angle of ascent = 60° Angle of descent 90° Angle of dwell between ascent and descent = 45° Speed of the cam = 200 rpm. Draw the profile of the cam and determine the maximum velocity and the uniform acceleration of the follower during the outstroke and the return stroke.	12	L4	5	1,2,3
OR					
VII.	A conical pivot supports a load of 20 kN, the cone angle is 120° and the intensity of normal pressure is not to exceed 0.3 N/mm^2. The external diameter is twice the internal diameter. Find the outer and inner radii of the bearing surface. If the shaft rotates at 200 r.p.m. and the coefficient of friction is 0.1, find the power absorbed in friction. Assume uniform pressure condition.	12	L3	3	1,2,3
OR					
VIII.	(a) What is a clutch? Name different types of clutch. (b) A single-plate clutch transmits 25 kW at 900 rpm. The maximum pressure intensity between the plates is 85 kN/m^2. The outer diameter of the plate is 360 mm. Both the sides of the plate are effective and the coefficient of friction is 0.25. Determine the (i) inner diameter of the plate (ii) axial force to engage the clutch.	4 8	L1 L3	3 3	1,2,3 1,2,3
OR					
IX.	(a) Derive Freudenstein's equation for a four bar mechanism. (b) Explain the overlay method in the synthesis of mechanisms.	8 4	L2 L2	3 3	1,2,3 1,2,3

Blooms's Taxonomy Levels

L1 - 23.33%, L2 - 18.33%, L3 - 38.33%, L4 - 20%.
